*Suggested approaches:*

- Use diagnostic assessments to determine reading levels and instructional priorities.
- Incorporate authentic documents into assessment, just as you would for instruction. Participants are more likely to see the value of an assessment when they can see how it relates to the skills they need on the job.
- Consider the impact of low digital skills in the administration and scoring of online reading assessments. Take steps to support participants with online assessments or use paper-based assessments where possible.
- Incorporate a range of complexity levels so that you can capture the upper and lower ranges of performance. Where possible, use adaptive assessments to adjust complexity level of questions based on performance, ensuring an appropriate level of challenge.

*Proficiency Levels:*

Proficiency levels are the level at which a person demonstrates a particular skill. These levels help practitioners build assessment tools to measure participants' skills. The Office of Skills for Success will refine the proficiency levels over time.

The following descriptors summarize the original 1-to-5 level divisions of the *Reader's Guide to Essential Skills Profiles* (2010). These descriptors are not aligned with the updated definitions and constructs of Skills for Success. More work is needed to tighten the alignment.

**Preliminary Proficiency Levels**

**Level 1:** Read relatively short texts so you can locate a single piece of information; follow simple written directions.

**Level 2:** Read more complex texts so you can locate a single piece of information or read simpler texts to locate multiple pieces of information; make low-level inferences.

**Level 3:** Choose and integrate information from various sources or from several parts of a single text; make low-level inferences from multiple sources.

**Level 4:** Integrate and synthesize information from multiple sources or from complex and lengthy texts; make complex inferences and use general background knowledge; evaluate quality of text.

**Level 5:** Interpret dense and complex texts; make high-level inferences and use specialized knowledge.

*Research Report to Support the Launch of Skills for Success (p. 44)*

**RESOURCES:**


- Curtis M. E., & Kruidenier J. R. (2005). *Teaching adults to read: A summary of scientifically based principles.* National Institute for Literacy. https://lincs.ed.gov/publications/pdf/teach_adults.pdf

- Degener, S., Purcell-gates, V., & Jacobson, E. (2003). *Creating authentic materials and activities for the adult literacy classroom: A Handbook for practitioners.* National Center for the Study of Adult Learning and Literacy. https://www.researchgate.net/publication/234717345_Creating_Authentic_Materials_and_Activities_for_the_Adult_Literacy_Classroom_A_Handbook_for_Practitioners

- Evetts, J., & Fownes, L. (1999). *Document use at work.* SkillPlan. https://skillplan.ca/learn/product/document-use-at-work/

- Government of Canada. (n.d.). *Reading indicator: A guide for employers.* https://www.canada.ca/en/services/jobs/training/initiatives/skills-success/tools/reading-indicator.html

- Learner Variability Navigator. (n.d) *Foundational reading skills.* Digital Promise. https://lvp.digitalpromiseglobal.org/content-area/adult-learner/factors/foundational-reading-skills-adult-learner/summary

- Lew, J., & Hardt, M. D. (2011). *Controlling complexity: An introduction to question structure.* SkillPlan. https://skillplan.ca/learn/product/controlling-complexity/

- PBS Learning Media. (n.d.) *Reading instructional strategies: Resources for adult educators.* https://www.pbslearningmedia.org/collection/ristrat/




# GRAB AND GO TRAINING IDEAS

| Activity | Reading Components | Instructional Tips | Sector-Specific |
|---|---|---|---|
| Provide small groups with a variety of documents of a particular type (e.g. tables, lists, signs, entry forms, diagrams, manuals, etc.) Focus on structure and features: How is the information organized? What are the common features of this document type? Which features make it easier/ harder to find information? Then focus on purpose: Who is the intended audience? How would you use this document on the job? Each group presents their findings to the class. | 4. Understand and apply the information<br>• *Comprehend the main theme of the document(s)*<br>5. Evaluate the document(s)<br>• *Evaluate the purpose, tone, and structure of the document*<br>6. Reflect on the document(s)<br>• *Determine the intended audience of the document*<br>• *Where relevant, reflect on how the author is using evidence and/ or language to achieve a particular purpose* | Use authentic documents related to participants' career and learning goals.<br><br>Option: give each group the same documents OR assign a different document type to each group so they are tasked with teaching the class.<br><br>Provide instruction on strategies for navigating documents through the use of features and structures (SFS 3.1.6). | Gather a variety of sector-specific workplace documents and sort them by type.<br><br>Sample Electrician Documents: WHMIS labels, shift schedules, worksite procedures, accident report forms, parts lists, assembling drawings, blueprints, electrical code manual, safety notices and manuals, material safety data sheets, etc. |
| Provide participants with a manual or standard operating procedure and task-based questions to answer: "The equipment won't start. What should you do?", "How do you clean up the spill?", etc. Talk through the answers together, drawing attention to how participants used reading strategies, question structure and document features. | 1. Identify the task that requires you to read<br>• *Identify the goals and purposes of the reading activity*<br>2. Identify the information contained in the document(s)<br>• *Use methods such as scanning, skimming to identify key information* | Provide explicit instruction and step-by-step strategies on how to approach reading tasks.<br><br>Ask participants to talk you through their process so you can identify where they are struggling.<br><br>Model metacognition using think aloud step-by-step processes (SFS 2.5.6). | Contextualize manuals, standard operating procedures (SOPs) and other documents to the sector.<br><br>e.g., Furniture assemblers use SOPs to carry out edge-banding and measure and mark out materials within acceptable tolerances. |
| Task participants with a scenario that requires them to decide between two or three options, such as which equipment to purchase, which procedure to follow, etc. Provide them with the documents needed to compare and evaluate the options. Participants share their decision and rationale with the class. | 2. Identify the information contained in the document(s)<br>• *Look through multiple pieces of information contained in documents*<br>• *Pull out relevant information from documents*<br>4. Understand and apply the information<br>• *Compare and contrast information* | Use authentic tasks and documents related to the decisions participants will need to make on the job.<br><br>Discuss communication skills needed to effectively explain a decision (e.g., clarity, using examples and facts, considering audience, etc.). | Contextualize the decision scenario depending on the sector and role.<br><br>Sample scenario for Cleaners and Housekeeping Staff: You come across a greasy spill in the carpeted hallway. Compare the cleaning labels to determine which product and procedure is best suited for the task. |

# Writing

Writing is the ability to share information using written words, symbols and images.

## Why are Writing Skills Important?

The changing labour market and advances in technology require writing skills that are suitable for different situations and digital platforms. At work, we use writing skills to write memos, emails, or reports. Writing skills are also needed in daily life to fill out a credit card or job application. Knowing what to write, how much to write and in which style to write is important. Writing skills ensure your writing is suitable for your purpose, the intended reader and the context.

**Writing Components**

1. Identify the task that requires you to write
2. Plan the writing task
3. Use written words and phrases so you can achieve the purpose of the writing task
4. Choose the appropriate language and style for the writing task
5. Choose the appropriate format for the writing task
6. Review and revise your writing

*Research Report to Support the Launch for Skills for Success* (p. 17)

| Sector | Workplace Examples |
|---|---|
| Construction | Boilermakers complete hazard or near-miss report forms to record information about occurrences. Since these documents could be used in a court of law, clarity, detail and accuracy are important. |
| Manufacturing | Mechanical Assemblers and Inspectors complete a variety of forms, such as time tickets, work orders, inspection sheets and defect forms, to record information about quality control and the use of material and human resources in assembling mechanical products. |
| Tourism | Travel Agents write emails to clients and travel companies to provide information, itineraries and explain problems. |
| Education | Early Childhood Educators write notes and emails to co-workers and parents, e.g. write notes to parents informing them about forms to be completed. |

## Transitioning from Essential Skills to Skills for Success

The most significant change to Writing in the Skills for Success framework is that it now includes elements of Document Use, which was considered a separate skill in the Essential Skills model.

Document Use has been relocated and is now embedded in the skills of Writing, Reading and Numeracy. This update reflects the frequently integrated nature of these skills in work, learning and life contexts.

The following table shows how Document Use has been integrated within Writing. The Skills for Success column on the left identifies the components and subcomponents of Writing that include elements of Document Use (identified in red within the *Research Report*). The Essential Skills column on the right lists the Document Use tasks from the original model that are relevant to Writing. Practitioners can use these cues to plan where to integrate existing Document Use materials into Writing instruction.

| Document Use Integration | |
|---|---|
| **Skills for Success**<br>*Writing components and subcomponents that include some Document Use elements* | **Essential Skills**<br>*Writing-related Document Use tasks from the original model* |
| 2. Plan the writing task<br>• Identify the information to include in the writing task<br>• Determine how much you need to write and the level of details to include<br>• Gather information<br><br>3. Use written words and phrases so you can achieve the purpose of the writing task<br>• Produce legible handwritten text when needed (Note: handwritten text may become obsolete over time)<br><br>5. Choose the appropriate format for the writing task<br>• To enhance the presentation of information or ideas use charts, tables, graphs, etc.<br>• Use pre-determined or pre-formatted workplace documents (e.g., accident report forms, timesheets, memo boards)<br><br>*Research Report to Support the Launch of Skills for Success (p. 17)* | • complete forms by marking check boxes, recording numerical information or entering words, phrases, sentences or texts of a paragraph or more<br><br>• create tables, schedules and other table-like text<br><br>• enter information on tables, schedules or other table-like text<br><br>• create assembly drawings.<br><br>• create schematic drawings<br><br>• make sketches<br><br>*Reader's Guide to Essential Skills Profiles (p. 25)* |

Practitioners can also use the writing components to guide and structure how they deliver instruction. The writing components are process-oriented, identifying the key skills and steps required to complete a writing task. This is a helpful update since the Essential Skills framework did not provide processes for a learner or practitioner to follow.

## Evidence-Based Instructional Strategies

The following instructional strategies have been selected to highlight established or promising practices within the Skills for Success field and/or significant research that supports the instruction of writing skills.

### 1 Integrated Grammar

Compared to academic and language learning contexts, the importance of writing mechanics (e.g., grammar, sentence structure, spelling, and punctuation) is generally de-emphasized in Skills for Success training. These components are still important in that they help writers communicate a clear message and enable them to achieve the greater purpose of the writing task. However, grammar instruction is not typically the focus of Skills for Success training. This is supported by research that has shown that teaching grammar alone can decrease the quality of participants' writing (American Institute for Research [AIR], 2012).

*Instructional Strategies:*

- Focus first on meaningful, real-world writing tasks then integrate instruction of grammar, sentence structure, spelling and punctuation on an as-needed basis. Some writing tasks will lend themselves to specific grammar points more than others, just as some participants will need more support with mechanics than others.
- Ask participants to consider the importance of correct spelling, grammar and punctuation relative to the audience and context of the writing task. For instance, more formal higher-stakes writing tasks (e.g. published materials) require greater accuracy than informal day-to-day tasks (e.g. emails between coworkers).

**Learning Transfer**



Align learning content and delivery modes with unique learner needs, skill levels, objectives and contexts.

*Implementation Guidance (p. 104)*

- Provide participants with strategies and tools to proofread and correct their writing. Use checklists and templates to guide self and peer review.




**Practitioner Competency Framework**

- SFS 3.3.1 Provides instruction on elements of writing
- SFS 3.3.2 Provides instruction on the relationship between audience, tone and formality
- SFS 3.3.6 Provides instruction on strategies for editing, revising and proofreading

## 2 Context-Driven

Writing tasks are context-driven, in that the language, tone and style will vary greatly depending on the audience and purpose for writing. To improve their ability to employ an appropriate writing style, participants need practice responding to a variety of writing contexts (Bazerman, 2015).

*Instructional Strategies:*

- Relate writing activities to participants' intended career goals. Identify the kinds of writing that workers do on the job then replicate these writing tasks in training. Guide participants through the writing components and processes, tackling the writing task as a worker would, including choosing an appropriate language and style.
- Incorporate other Skills for Success as relevant, such as communication, collaboration and digital skills, helping participants to build multiple skills and apply them to authentic tasks.
- Use collaborative writing strategies to help participants gain greater understanding of the social dynamics at play in writing tasks. For example, ask participants to share their writing choices related to formality, tone, language, persuasive techniques, etc. Peer-feedback also provides participants with valuable information about how the audience has received and interpreted their message.

**Engagement**



Integrate multiple Skills for Success into programming model to reflect the interactive nature of skills.

*Implementation Guidance (p. 105)*




**Practitioner Competency Framework**

- SFS 2.2 Contextualizes Skills for Success training to make learning relevant and practical
- SFS 3.3.12 Integrates collaboration, communication and digital skills into writing tasks to help participants generate ideas, peer edit, revise writing and share knowledge

## 3 Writing Process

The components in the *Research Report to Support the Launch of Skills for Success* (2021) take a process-based approach to writing. Though not entirely sequential, the components identify the key skills and steps required to complete a writing task.

Practitioners need to provide explicit instruction on each step and give participants lots of practice and support through the writing process.

*Instructional Strategies:*

- Explicitly teach a set of steps for participants to follow each time they approach a writing task. Consistently reinforce the steps throughout the training program so that participants internalize them over time.
- Provide participants with ample opportunities to practice their writing. Reduce the marking load by using checklists, peer-feedback and targeted approaches to feedback (e.g., only commenting on select aspects of the writing).
- Provide models and templates for participants to follow. For instance, give participants a completed entry form and a blank form to fill out for a different context. Peer-created models of writing can be particularly effective for illuminating different approaches to a writing task.

**Note**: See Approach #4 Section 2 for more information about Direct Instruction.




**Practitioner Competency Framework**

- SFS 2.6.3 Guides the learning process with step-by-step processes, templates, demonstrations and/or models for participants to follow
- SFS 3.3.3 Provides instruction on organizational strategies to help participants approach writing tasks systematically
- SFS 3.3.6 Provides instruction on strategies for editing, revising and proofreading

## Assessment Considerations

The assessment of writing skills most commonly involves the scoring of a writing sample against various criteria.

*Suggested approaches:*

- Use scoring rubrics with detailed dimensions and criteria
  - Checklist ratings may allow for better standardization
  - Holistic ratings capture overall feeling and nuance
- Ensure marking is reliable through consensus and/or consistency approaches
- Allow participants to handwrite or type depending on their preferences
- Incorporate frequent formative assessments to provide constructive, timely feedback and enable participants to learn from their mistakes

*Proficiency Levels:*

Proficiency levels are the level at which a person demonstrates a particular skill. These levels help practitioners build assessment tools to measure participants' skills. The Office of Skills for Success will refine the proficiency levels over time.

The following descriptors summarize the original 1-to-5 level divisions of the *Reader's Guide to Essential Skills Profiles* (2010). These descriptors are not aligned with the updated definitions and constructs of Skills for Success. More work is needed to tighten the alignment.

**Preliminary Proficiency Levels**

**Level 1:** Write less than a paragraph to organize, remind, or inform.

**Level 2:** Write brief text that is a paragraph or longer, to serve a variety of purposes. Content of writing is routine, with little variation from one instance to the next.

**Level 3:** Write either longer or shorter pieces to inform, explain, request information, express opinions or give directions.

**Level 4:** Write longer pieces, which present considerable information and which may feature a comparison or analysis.

**Level 5:** Write pieces of any length which demand originality and effectiveness. This includes creative writing. Appropriate tone and mood may be as important as the content.

*Research Report to Support the Launch for Skills for Success (p. 45)*

**RESOURCES:**


- ABC Literacy (2023). *Writing.* UP Skills for Work. https://upskillsforwork.ca/wp-content/uploads/ABC-UP-Writing.pdf
- U.S. Department of Education, Office of Vocational and Adult Education. (2011). *Just Write! Guide.* https://lincs.ed.gov/sites/default/files/TEAL_JustWriteGuide.pdf
- Bow Valley College. (2016). *Writeforward.* http://www.writeforward.ca/
- MacArthur, C. A., Graham, S., & Fitzgerald, J. (Eds.). (2015). *Handbook of writing research* (pp. 11–23). Guilford Press.
- McLean, E., & Griffiths, K. (2022). *Writing and writing instruction.* Australian Education Research Organization. https://www.edresearch.edu.au/resources/writing-and-writing-instruction

## GRAB AND GO TRAINING IDEAS

| Activity | Writing Components | Instructional Tips | Sector-Specific |
|---|---|---|---|
| Scaffold participants' email writing skills by providing a model to analyze and follow. Label the parts of the email. Discuss word choice and tone in relation to the audience. Have participants correct an example of an email that is too formal or informal. Now participants are ready to write their own emails in response to workplace scenarios. | 1. Identify the task that requires you to write Identify the goals and purposes of the writing task<br>• *Identify the audience*<br><br>4. Choose the appropriate language and style for the writing task<br>• *Use formal and informal tone and language appropriate for audience and context* | Determine if participants need more practice on formal or informal styles and teach to that.<br><br>Provide opportunities for peer review to enable participants to learn from each other's writing choices. | Generate sector-specific scenarios based on the typical occupational email writing tasks.<br>  <br>e.g., Spa managers write emails to staff to communicate schedule changes, to clients to answer service-related questions, and to manufactures to express concerns about products. |
| Teach participants how to summarize an event to complete an incident report form. Discuss strategies for summarizing (e.g., using the 5 Ws). Participants can summarize an incident or accident they have observed or report about a video simulation (see provincial OH&S websites). Integrate problem solving by discussing ways to prevent future incidents. | 2. Plan the writing task<br>• *Identify the information to include in the writing task*<br>• *Determine how much you need to write and the level of details to include*<br><br>5. Choose the appropriate format for the writing task<br>• *Use pre-determined or pre-formatted workplace documents* | Define purpose and explain the purpose of different types of entry forms.<br><br>Define primary and secondary information.<br><br>Demonstrate how to respond to a variety of response modes (checkboxes, words, phrases, sentences) found in entry forms. | Use authentic incident report forms from sector-specific companies or occupational health and safety (OH&S) organizations.<br>  <br>Gather written case studies or video simulations of accidents or near misses in the construction industry for participants to summarize. |
| After reviewing several health and safety or standard operating procedures (SOPs), ask participants to write their own procedural document. Provide clear criteria: e.g., present tense, plain language, bullet points, visuals, etc. Extend the activity by integrating communication skills: participants take on the role of supervisor to explain a new procedure to their team. | 5. Choose the appropriate format for the writing task<br>• *To organize information or ideas, use paragraphs, bullet points, numbered list, sub-headings, etc.*<br>• *Use pre-determined or pre-formatted workplace documents*<br><br>6. Review and revise your writing<br>• *Proofread and revise your writing for accuracy, meaning and tone* | Define process and procedure and the role they play in the workplace.<br><br>Analyze workplace documents with participants to identify common elements of a process/procedure, such as present tense, plain language, bullet points, visuals, etc. | Contextualize the health and safety or standard operating procedures (SOPs) depending on the sector and role.<br>  <br>e.g., Manufacturing technicians use SOPs to operate equipment, calibrate equipment, dispose of hazardous waste, conduct quality testing and perform maintenance. |

# Numeracy

Numeracy is the ability to find, understand, use and report mathematical information presented through words, numbers, symbols and graphics.

## Why are Numeracy Skills Important?

The modern economy requires numeracy skills that go beyond basic arithmetic, and understanding numbers remains critical to functioning in today's society. Many jobs require the ability to work with numbers and math. For example, we use numeracy skills to measure materials or count inventory at work. Numeracy skills are also needed in a wide variety of daily contexts. For example, you use numeracy skills to manage your finances or to make sense of statistics in the news.

**Numeracy Components**

1. Identify the task that requires you to use numeracy
2. Identify the mathematical information
3. Make connections between related pieces of mathematical information
4. Apply mathematical operations and tools you will need to answer the question
5. Interpret and evaluate the information
6. Share the mathematical information, results and implications

*Research Report to Support the Launch of Skills for Success (p. 19)*

| Sector | Workplace Examples |
|---|---|
| Construction | Electricians calculate offsets to allow for bends around obstacles by using indirect measurement, solving square roots or by using trigonometric constants to construct standard bends in conduit. |
| Manufacturing | Machine Operators in mineral processing use a tape measure to measure hot and cold bars to determine shrinkage. They have to be aware of the correct "hot" dimensions which will shrink to the correct cold measure. |
| Food Processing | Fish plant workers calculate the weight of fish to send down the production line so the end of the line receives the correct weights, considering percentages of waste. |
| Health Care | Dental hygienists estimate the time required to complete dental procedures, based on patient needs and their own experience conducting similar procedure. |

## Transitioning from Essential Skills to Skills for Success

The most significant change to Numeracy in the Skills for Success framework is that it now includes elements of Document Use, which was considered a separate skill in the Essential Skills model.

Document Use has been relocated and is now embedded in the skills of Reading, Writing and Numeracy. This update reflects the frequently integrated nature of these skills in work, learning and life contexts.

The following table shows how Document Use has been integrated within Numeracy. The Skills for Success column on the left identifies the components and subcomponents of Numeracy that include elements of Document Use (identified in red within the *Research Report*). The Essential Skills column on the right lists the Document Use tasks from the original model that are relevant to Numeracy. Practitioners can use these cues to plan where to integrate existing Document Use materials into Numeracy instruction.

| Document Use Integration | |
|---|---|
| **Skills for Success**<br>*Numeracy components and subcomponents that include some Document Use elements* | **Essential Skills**<br>*Numeracy-related Document Use tasks from the original model* |
| 2. Identify the mathematical information<br>• Locate key details, concepts, and other mathematical information<br>• Use methods such as scanning and skimming to identify relevant information<br><br>3. Make connections between related pieces of mathematical information<br>• Make connections between known and unknown in the mathematical problem<br>• Make connections between different parts of the information presented<br><br>4. Apply mathematical operations and tools you will need to answer the question<br>• Order or sort<br><br>6. Apply mathematical operations and tools you will need to answer the question<br>• Use different means and methods to share the information, its results, and implications, such as in a presentation, in writing, through a diagram, map or graph.<br><br>*Research Report to Support the Launch of Skills for Success (p. 19)* | • complete forms by marking check boxes, recording numerical information or entering words, phrases, sentences or texts of a paragraph or more<br>• read completed forms containing check boxes, numerical entries, phrases, addresses, sentences or texts of a paragraph or more<br>• read tables, schedules and other table-like text<br>• plot information on graphs (e.g., line, pie, bar)<br>• obtain specific information from graphs or charts<br>• interpret information on graphs or charts<br>• construct or draw graphs or charts.<br>• recognize common angles such as 15˚, 30˚, 45˚ and 90<br>• draw, sketch or form common shapes such as circles, triangles, spheres, rectangles, squares, etc.<br>• take measurements from scale drawings<br>• draw to scale<br><br>*Reader's Guide to Essential Skills Profiles (p. 25)* |

Practitioners can also use the numeracy components to guide and structure how they deliver instruction. The numeracy components are process-oriented, identifying the key skills and steps required to complete a numeracy task. This is a helpful update since the Essential Skills framework did not provide processes for a learner or practitioner to follow.

## Evidence-Based Instructional Strategies

The following instructional strategies have been selected to highlight established or promising practices within the Skills for Success field and/or significant research that supports the instruction of numeracy skills.

### 1 Learner Attitudes and Emotions Relating to Numeracy

Negative beliefs about ability and emotions about math can interfere with success in adult numeracy instruction. Adult learners who have had challenges with math need support to develop a sense of competence and confidence in learning mathematics and numeracy. Facilitate safe learning spaces to allow learners to focus on learning, develop skills and relationships, and build the confidence necessary to succeed.

*Instructional Strategies:*

- Create instructional environments where math anxiety can be openly discussed. Build on prior knowledge and strengths, and pair with opportunities for success to allow learners to reflect on their growth throughout a program.
- Help learners build reading skills that support acquisition of mathematical language and notation. Model the creation of glossaries for math terminology so they can translate tasks into numeracy operations with confidence.
- Scaffold learning in numeracy; build and emphasize success in the foundational skills to support the learning of more complex concepts and skills.
- Adjust teaching styles and implement the principles of Universal Design for Learning to address learning differences that may influence accessibility and anxiety about numeracy instruction.



Under represented Groups

**Engagement**



Create safe learning spaces by leveraging the social emotional Skills for Success to build learners' self-efficacy, autonomy, and confidence.

*Implementation Guidance (p. 109)*




**Practitioner Competency Framework**

- SFS 2.8 Supports social-emotional well-being to develop self-regulation skills
- SFS 2.9 Applies a learner-centred approach to Skills for Success training
- SFS 3.2.4 Provides instruction on the language of numeracy
- SFS 3.2.7 Recognizes, assesses and addresses participants' diverse and potentially negative experiences, beliefs and emotions towards numeracy that may interfere with learning

## 2 Contextualized Numeracy Instruction

Participants in Skills for Success training programs are motivated to learn new skills when math problem-solving is connected to their interests as well as goal-oriented contexts. With a wide variety of educational backgrounds and experience, learners may not have the foundational numeracy skills they need to complete numeracy tasks in training and on the job. Therefore, teach (or reteach) foundational skills in contextualized, realistic applications that connect to what they already know and what they want to know. Contextualization promotes conceptual understanding and facilitates the scaffolding of learning up to new, abstract concepts.

*Instructional Strategies:*

- Learn about your participants and connect instruction back to their interests and prior life experiences to engage interest, as well as tie it forward to their goals (Ginsburg et al., 2011, p. 31). By working with pre-existing knowledge, skills and strategies, new knowledge and skills are more easily learned, understood and applied (Dingwall, 2000, pp. 20-22).
- Situate math instruction in the real world. Use everyday or training-related manipulatives and activities to transfer learning between math in theory and math in practice (Ginsburg & Gal, 2000). Use calculators and digital technology that will also be used in further training and the workplace (Ginsburg et al., 2011). Improve numeracy skills by providing participants with the opportunity to practice many times with new workplace scenarios. Research shows that using a range of contexts in math instruction was most effective in improving skills in new problem-solving applications (Ginsburg et al., 2011, p. 31).
- Pace instruction to the skill level of your learners and guide their reflection of personal skill development within the framework of the training program. Offer frequent opportunities for self-assessment and reflection on effort and results. Point out their individual achievement in reaching milestones in overall training goals.

**Engagement**



Align learning content and delivery modes with unique learner needs, objectives, skill levels and contexts.

*Implementation Guidance (p. 104)*

**Note**: See Approach #2 in Section 2 to learn more about contextualizing instruction across skills, sectors and audiences.




**Practitioner Competency Framework**

- SFS 3.2.5 incorporates concrete, visual and hands-on activities to develop participants' understanding of numeracy concepts
- SFS 3.2.6 Provides participants with opportunities to reflect on their prior learning in numeracy and connect it to new concepts
- SFS 3.2.8 Integrates authentic documents into numeracy activities, including culturally and contextually relevant documents

## 3 Collaborative Learning

Collaborative learning refers to interactive instructional strategies, which can involve formal and more casual approaches. It can positively influence mathematics performance and be of significant benefit to marginalized students such as those from low-income backgrounds (Ginsburg et al., 2011).

*Instructional Strategies:*

- Implement various learner grouping strategies (homogenous skill level, peer tutor, student choice, interest, etc.) to enhance learning through communication and collaboration (Coben et al., 2003). Encourage students in peer learning groups to offer their perspectives on how they visualize or break down math problems. Teaching each other can also help learners process their skills in a deeper manner.
- Provide a clear structure of strategies and activities for successful collaborative learning, whether with an instructor or with learner groups. Collaborative learning helps prepare adults for communication demands in the workplace and allows for learning between learners from diverse cultural backgrounds (Ginsburg et al., 2011).
- Tie in instruction on Communication skills. Research identifies that communication plays a critical role in numeracy learning. It helps learners develop problem-solving skills, integrate new information into previous knowledge, and develop and expand on their own thinking and the ideas of others by engaging in discussion (AIR, 2014). Aspects of language acquisition may also develop when supplemented with conceptual tasks and activities that focus on mathematics (Coben et al., 2003).

**Engagement**

Integrate multiple Skills for Success into programming model to reflect the interactive nature of skills.

*Implementation Guidance (p. 105)*




**Practitioner Competency Framework**

- SFS 3.2.3 Provides participants with opportunities to use different numeracy approaches and strategies and to explain their reasoning
- SFS 3.6.10 Integrates discussion, brainstorming and collaboration into problem-solving activities
- SFS 3.8.7 Creates learning tasks that require participants to consider their own and others' perspectives, biases, purposes and intentions

## Assessment Considerations

Numeracy assessments typically require participants to complete mathematics tasks of varying complexity. Correct and incorrect answers can be scored objectively. There is a rich body of national and international research to support the creation of numeracy assessments with demonstrated reliability and validity (e.g., PIACC and TOWES).

Instruction of numeracy skills should include formative assessment with constructive feedback which is descriptive, specific, timely and measured. This allows learners to understand their mistakes and adjust their thinking and processes so they can reach appropriate solutions (AIR, 2014). A balance of both formative and summative assessments is important for gathering information about learners' knowledge and understanding of numeracy concepts (Ginsburg et al., 2011).

*Suggested approaches:*

- Use diagnostic assessments to determine numeracy levels and instructional priorities.
- Incorporate authentic tasks into assessment, just as you would for instruction. Participants are more likely to see the value of an assessment when they can see how it relates to the skills they need on the job.
- Consider the impact of low digital skills in the administration and scoring of online numeracy assessments. Take steps to support participants with online assessments or use paper-based assessments where possible.
- Incorporate a range of complexity levels so that you can capture the upper and lower ranges of performance. Where possible, use adaptive assessments to adjust complexity level of questions based on performance, ensuring an appropriate level of challenge.
- Build persistence and confidence by involving learners in assessment and recognizing their progress towards goals (Ginsburg et al., 2011).
- Research recommends that practitioners approach assessment in a way that is cognizant of their learners' math anxieties and that helps learners understand and correct their mistakes. Apply the principles of Universal Design for Learning to address learning differences and offer alternate modes of assessment to demonstrate learning.
- Offer opportunities to retake formative assessments to measure mastery and also highlight individual progress.

*Proficiency Levels:*

Proficiency levels are the level at which a person demonstrates a particular skill. These levels help practitioners build assessment tools to measure participants' skills. The Office of Skills for Success will refine the proficiency levels over time.

The following descriptors summarize the original 1-to-5 level divisions of the *Reader's Guide to Essential Skills Profiles* (2010). These descriptors are not aligned with the updated definitions and constructs of Skills for Success. More work is needed to tighten the alignment.

**Preliminary Proficiency Levels**

**Level 1:**

- *Operations Required*: Only the simplest operations are required and the operations to be used are clearly specified. Only one type of mathematical operation is used in a task.
- *Translation*: Only minimal translation is required to turn the task into a mathematical operation. All information required is provided.

**Level 2:**

- *Operations Required*: Only relatively simple operations are required. The specific operations to be performed may not be clearly specified. Tasks involve one or two types of mathematical operation. Few steps of calculations are required.
- *Translation*: Some translation may be required or the numbers needed for the solution may need to be collected from several sources. Simple formulae may be used.

**Level 3:**

- *Operations Required*: Tasks may require a combination of operations or multiple applications of a single operation. Several steps of calculation are required.
- *Translation*: Some translation is required but the problem is well defined. Combinations of formulae may be used.

**Level 4:**

- *Operations Required*: Tasks involved multiple steps of calculation.
- *Translation*: Considerable translation is required.

**Level 5:**

- *Operations Required*: Tasks involve multiple steps of calculation. Advanced mathematical techniques may be required.
- *Translation*: Numbers needed for calculations may need to be derived or estimated; approximations may need to be created in cases of uncertainty and ambiguity. Complex formulae, equations or functions may be used.

*Research Report to Support the Launch of Skills for Success (p. 45)*

**RESOURCES:**


- ABC Literacy. (2023). Numeracy. UP Skills for Work. https://upskillsforwork.ca/wp-content/uploads/ABC-UP-Numeracy.pdf
- U.S. Department of Education, Office of Career, Technical, and Adult Education. (2014). *Math Works! Guide*. https://lincs.ed.gov/sites/default/files/Teal_Math_Works_Guide_508.pdf
- SkillPlan. (2023). *Learning hub*. https://skillplan.ca/learn/
- Skills/Compétences Canada. (2022). Workbook: Numeracy. Skills for Success. https://yourskillsforsuccess.com/workbook/numeracy/




# GRAB AND GO TRAINING IDEAS

| Activity | Numeracy Components | Instructional Tips | Sector-Specific |
|---|---|---|---|
| Provide pairs with an imperial tape measure, and a list of trade and classroom items to measure in inches and fractions of an inch. The items should vary and challenge the learners to figure out how to use the tool.<br><br>Focus on strategies for reading with precision. Debrief on the measurements, strategies and working together. | 2. Identify the mathematical information<br>• *Make connections between known and unknown in the mathematical problem*<br><br>4. Apply mathematical operations and tools you will need to answer the question<br>• *Measure*<br>• *Apply a combination of operations and tools to complete a task* | Teach math foundations related to fractions, lowest common denominator, simplifying, reading a small ruler first.<br><br>This is a good activity for collaborative learning.<br><br>Encourage a growth mindset by praising and commenting on participants' efforts and processes rather than only doing things one way (SFS 3.7.5). | Gather a variety of sector-specific items to measure.<br><br>Sample Piping items: lengths of pipe and tube, fittings with different diameters, ladders, workbench, classroom ceiling height and floor dimensions, students' heights, inside window dimensions, diagonal doorway dimension, etc. . |
| Provide a variety of contextualized materials and typical dimensions for the occupation.<br><br>Ask learners to calculate the volume and the weight of simple shapes of items.<br><br>Discuss answers, drawing attention to how participants used a strategy to break down each question. | 2. Identify the mathematical information<br>• *Make connections between known and unknown in the mathematical problem*<br><br>5. Interpret and evaluate the information<br>• *Evaluate the information or results within the context and whether they make sense* | Model metacognition using think aloud step-by-step processes (SFS 2.5.6) and how to approach numeracy tasks.<br><br>Build complexity from rectangular items with whole-number dimensions up to complex shapes. Ask participants to talk you through their process so you can identify where they are struggling. | Contextualized list of items with typical dimensions.<br><br>e.g., Truck drivers calculate weights of building materials such as masonry units, lumber, drums of oil, drywall sheets, steel pipes. Loads can be of mixed materials. |
| Provide a table of data with item subsets that can be grouped and compared.<br><br>Provide a scenario that requires presenting information in a pie chart or bar graph. Participants must calculate percentages and decide how to display the percentages on a graph. | 1. Identify the task that will require you to use numeracy<br>• Identify the form of the response expected<br><br>6. Share the mathematical information, results, and implications<br>• *Use different means and methods to share the information, its results, and implications, such as in a presentation, in writing, through a diagram, map or graph* | Model the reading skills required to gather data for each demographic. Model alternate methods for calculating percents, step-by-step. Encourage learners to try both methods.<br><br>Building bar graphs may be easier than pie charts. Provide the gradation for the y-axis. Use graph paper.<br><br>This activity can integrate Digital skills: you can teach how to use common software (Word, Excel) to create graphic displays. | Contextualize the data analysis depending on the sector and role.<br><br>Sample scenario for Heritage Interpreters: Create a graph showing the percentage breakdown of attendance by demographic groups in the previous month. |

# Digital

Digital is your ability to use digital technology and tools to find, manage, apply, create and share information and content.

## Why are Digital Skills Important?

Digital technology has changed the way you find and share information, solve problems, and communicate with others. Most jobs now use digital skills, and you need them when you apply other skills such as reading, writing and numeracy.

Digital skills help you keep up with changing demands in the modern workplace. In daily life, you need digital skills to connect safely socially and to make use of online resources and services.

**Digital Components**

1. Use digital devices including computers, tablets, smart phones and other handheld devices
2. Use common digital tools to complete tasks
3. Use digital information
4. Use online tools and platforms
5. Apply safe and responsible practices online
6. Update and upgrade digital skills

*Research Report to Support the Launch of Skills for Success (p. 21)*

| Sector | Workplace Examples |
|---|---|
| Construction | Sheet Metal Workers use computer-assisted design (CAD) programs to create elevation, plan and sectional views of sheet metal fabrications. They may use augmented reality (AR) to see how their work overlaps with the work of other trades. |
| Manufacturing | Shippers and Receivers use digital devices to access and record shipment information, enter load weights, print orders and co-ordinate unloading schedules. |
| Tourism | Special Events Coordinators use graphic design applications to prepare advertising layouts, notices and programs. They may use event planning software and critical path software. |
| Health Care | Health Care Assistants use databases to retrieve and update patient information. For example, they use databases to enter data and search patient records for information about test results, treatments and medical conditions. |

## Transitioning from Essential Skills to Skills for Success

The Essential Skills Framework included the skill of Computer Use. In Skills for Success, it has been broadened to Digital, which includes the use of computers as well as other digital devices and platforms.

Digital has been updated to incorporate new technology and applications in the labour market. It now reflects a wide range of basic and advanced digital skills needed in today's world of work, including the ability to apply safe and responsible practices online.

Practitioners can update their existing Essential Skills training materials by incorporating current digital devices and tools and adding online safety components. To reflect the increasingly digital workplace, practitioners should integrate digital skills into their instruction of all Skills for Success.

## Evidence-Based Instructional Strategies

The following instructional strategies have been selected to highlight established or promising practices within the Skills for Success field and/or significant research that supports the instruction of digital skills.

### 1 Needs Assessment

Conducting a needs assessment is a good starting point for any training program, and the research shows this practice may be particularly important for digital skills. Participants in digital literacy courses often lack common backgrounds and may have diverse learning needs (Delfino et al., 2008). A thorough needs assessment is critical for addressing participants' diverse needs, skills and abilities.



Under represented Groups

*Instructional Strategies:*

- Use diagnostic tools to identify participants' backgrounds, goals, interests, skills and abilities. Be mindful to provide a supportive environment as some participants come with a lack of experience and confidence in their digital skills.
- Take a flexible approach when designing and delivering training to help ensure appropriate starting points are established for each participant. Where possible, involve participants in the instructional planning process to ensure programming meets their needs.
- Incorporate participants' prior experience, interests and intended career goals into course design to maximize relevance and engagement.

**Assessment**



Use Prior Learning Assessment & Recognition (PLAR) to document Skills for Success developed in other contexts

*Implementation Guidance (p. 123)*




**Practitioner Competency Framework**

- SFS 3.9.2 Differentiates instruction and curriculum to address participants' different learning needs and starting points
- DESIGN 2.2.1 Develops flexible curriculum materials and activities that can be adjusted to individual needs
- DELIVER 2.2 Recognizes and builds upon participants' prior learning, knowledge and experience
- ASSESS 1.2.1 Administers a pre-assessment to measure participants' levels, skills, abilities, goals, background knowledge and pre-requisites

## 2 Addressing Confidence

Research shows that lack of confidence is a significant barrier to improving digital skills. Older participants and participants from marginalized groups may be particularly affected (Moss, 2021; see also McGillivray et al., 2017). The following strategies can be used to help shape a supportive learning environment and build participant confidence in their digital skills.

Under represented Groups

*Instructional Strategies:*

- Build relationships with your participants, taking a friendly and informal approach. Provide opportunities for participants to get to know each other to cultivate a sense of community in the training room.
- Provide one-to-one support. Where possible, have volunteers or tutors monitor the computer lab and provide "just-in-time" coaching when a participant encounters a challenge.
- Start with simple activities then build up the complexity of digital tasks. Vary your instruction depending on participants' abilities and needs. Taking a gradual approach to new concepts helps avoid cognitive overload.

**Engagement**

Use a relationship-centred approach to engage learners.

*Implementation Guidance (p. 109)*




**Practitioner Competency Framework**

- SFS 2.5.2 Guides participants through progressively more complex learning tasks
- SFS 3.9.3 Addresses the needs of participants with low confidence in their digital skills
- DELIVER 1.2 Creates an atmosphere that addresses the social-emotional needs of participants

# 3 Problem-Based Approach

In problem-based learning, practitioners provide participants with tasks, scenarios, or challenges to solve. This approach has been found to be an effective way to deliver digital skills instruction. Participants learn by doing and seeing how their digital skills relate to real world tasks.

*Instructional Strategies:*

- Contextualize problem-based tasks to participants' career goals and interests. Participants are more motivated to learn digital skills when they can see real-world applicability. Increase transfer of learning to new digital technologies by helping participants to see the big picture context of a digital task – the "when" and "why" – not just the technical "how".
- Provide explicit instruction to support problem-based learning. A purely trial and error approach can be frustrating for participants. Instead, explain and model the process for solving technology-based problems.
- Develop participant autonomy by offering self-directed learning models. Provide opportunities for participants to move through material at different rates. Cultivating the ability to learn independently is critical to future-proofing participants' digital skills in the face of constant change in digital technologies.

**Learning Transfer**

Develop teaching tools and resources that support continued application of Skills for Success in new learning and working contexts.

*Implementation Guidance (p. 133)*




**Practitioner Competency Framework**

- SFS 2.2.2 Creates task-based learning activities based on real-world, workplace applications
- SFS 3.6.2 Uses problem-solving models that provide systematic processes for participants to follow
- SFS 3.9.1 Provides instruction on the technical "how to" of digital tasks and the contextual "when" and "why" to foster participant transfer of learning to new digital technologies

## Assessment Considerations

Digital skills are often assessed based on people's perception of their own skills, rather than systematic or standardized assessments that are more objective in nature. The Research Report to *Support the Launch of Skills for Success* (2021) outlines the pros and cons of several methods, such as task-based assessments, situational judgement tests, and subjective reports (see Table 2, p. 62).

*Suggested approaches:*

- Provide frequent formative feedback as your monitor participants working on a digital tasks.
- Incorporate structured peer feedback. Guide the feedback process by providing a process to follow.
- Create rubrics to help participants self-assess and monitor their learning.

*Proficiency Levels:*

Proficiency levels are the level at which a person demonstrates a particular skill. These levels help practitioners build assessment tools to measure participant's skills. The Office of Skills for Success will refine the proficiency levels over time.

**Preliminary Proficiency Levels**

**Entry:** You can use basic functions of familiar digital devices. You need guidance to find and evaluate the relevance and reliability of online information, and to engage in safe online practices.

**Intermediate:** You can use a wider range of functions of familiar and unfamiliar digital devices, including customizing devices for specific purposes (e.g., download and use an app, set up macros to automate tasks). You can find and use relevant and reliable online information and engage in safe online practices.

**Advanced:** You have in-depth knowledge of digital device operations and information technology systems. You can find and use relevant and reliable online information to improve digital processes, including enhancing your own online safety. You can assess future digital needs and keep your own digital skills up to date.

*Research Report to Support the Launch of Skills for Success (p. 46)*